Problem 1. Completely read the Homework Guidelines that are posted on Canvas.

Problem 2. Let x be an even integer and let n be a natural number. Prove that x^n is an even integer.

Problem 3. Let $n \in \mathbb{N}$. (In other words, n is a natural number.)

1. Make at least two conjectures about factorizations/primeness of $3^n - 1$.
2. Make at least two conjectures about factorizations/primeness of $3^n - 2^n$.

For each of these, you can make a table like in the example we did on the first day of class to guide your conjectures. You do not need to prove your conjectures.

Note that the `enumerate` environment creates a numbered list.
In the preamble, I created a `solution` environment with a $\square$ symbol.

Solution.

1. Here is my answer to part 1. We have that $\int 2x \, dx = x^2$. Or perhaps I want an equation on its own line:
$$\sin^2(x) + \cos^2(x) = 1.$$
Maybe I want to *emphasize* some text.
2. Maybe the next requires a proof. I created a command which invokes the `proof` environment built into one of our packages. It includes a box at the end of it. Here is my proof for part (b).

$\square$

Google is your best friend for LaTeX questions, but I'm also happy to answer any questions you have! In the notes and in lecture, I will try to point out any new commands I am using.